Arzida Binta Anwar

Dhaka, Bangladesh

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EDUCATION

Bangladesh University of Engineering & Technology (BUET) May 2025 – Jan 2027 (exp.)
M.Sc. in Civil Engineering (Major: Geotechnical Engineering) **CGPA: 3.17/4.00**

Bangladesh University of Engineering & Technology (BUET) Apr 2019 – Nov 2024
B.Sc. in Civil Engineering (Major: Geotechnical Engineering) CGPA: 2.80/4.00 (Last Sem.: 3.15/4.00)

- **Relevant Courses:** Soil Mechanics · Earthquake Engineering · Constitutive Modelling in Soil Mechanics · Earth Dams and Stability of Slopes · Reinforced Earth · Foundation Engineering · Earth Retaining Structures · Soil–Water Interaction

RESEARCH INTERESTS

Sustainable Ground Improvement • Seismic Soil Behavior & Liquefaction Mitigation • Durability and Seismic/ Environmental Resilience of Treated Geomaterials • Soil Dynamics, Site Response & Numerical Modeling

PUBLICATIONS

[1] S. Paul, P. Chakraborty, **A. B. Anwar**, G. Mortoza, and L. S. Sukti, **"Effect of Cement and Sugarcane Bagasse Ash on the Micro-Mechanical Characteristics of Expansive Soil,"** *Results in Engineering*, 106874, 2025. **(Q1, IF: 7.9, CiteScore: 7.3)** DOI: 10.1016/j.rineng.2025.106874

[2] S. Paul, A. B. Faruque, A. Mottaqi, and **A. B. Anwar**, **"Enhancing Shear Strength of Sandy Soils Using Enzyme-Induced Carbonate Precipitation and Sodium Alginate Biopolymer,"** *Geo-Congress 2026.* DOI: 10.1061/9780784486764.037

[3] S. Paul, **A. B. Anwar**, P. Chakraborty, and M. F. Alam, **"Evaluating In-Situ Soil Stiffness and Load-Settlement Behavior in Improved Subgrade Soil: Insights from Static Plate Load Tests,"** *7th International Conference on Advances in Civil Engineering (ICACE)*, 2024. Link

RESEARCH EXPERIENCE

Graduate Thesis, Dept. of CE, BUET May 2025 – Present
Supervisor: Prof. Mohammad Shariful Islam
Stabilization of Dispersive Soil using EICP (Enzyme-Induced Carbonate Precipitation) and Biopolymer
Developing a sustainable technique for stabilizing dispersive soils using EICP combined with biopolymers, focusing on improving soil strength, reducing dispersion potential, and enhancing durability through eco-friendly bio-mediated treatment.

Forensic Geotechnical Research, Dept. of CE, BUET Ongoing
Supervisor: Prof. Mohammed Kabirul Islam
Forensic Analysis of Road Embankment and RCC Retaining Wall Failure, Kashimpur, Gazipur
Numerical analysis of the 2026 collapse of a pile-supported RCC retaining wall and road embankment on soft clay along the Turag River using PLAXIS 2D.

- Identifying the failure mechanism and assessing the effects of pile spacing, soil strength, and traffic surcharge.
- Targeting publication in a forensic engineering journal.

Research Assistant, Dept. of CE, BUET June 2025 – Present

- **Bio-cementation of Expansive Soil via MICP – Large-Scale Overview** (ongoing): Tested physical and mineralogical properties of expansive soil, performed laboratory MICP trials on large-scale specimens, and preparing the manuscript.
- **Stabilization of Expansive Soil using EICP and Sewage Sludge Ash – Large-Scale Overview** (ongoing): Tested physical properties of expansive soil and conducted laboratory trials to find optimum additive content for large-scale effectiveness.
- **Vetiver-based Bioengineering for Slope Stabilization – Practical Guidelines for Bangladesh**: Developed engineering illustrations and schematic drawings, and contributed to design and implementation guidelines suited to local geotechnical conditions.
- **CSEB (Compressed Stabilized Earth Block) Manufacturing – From Lab to Field**: Analyzed soil physical and mechanical properties for CSEB production and compared laboratory-produced vs. factory-made blocks.

Research Assistant, Dept. of CE, BUET Aug 2025 – Oct 2025
Construction of Bridge over Feni River on Sonagazi & Mirsharai Economic Zone Connecting Road

Structural modeling and analysis of a 640 m progressive cantilever box-girder bridge in Midas Civil, developing a detailed construction-stage model incorporating tendon layout, prestressing sequences, and time-dependent material properties (creep, shrinkage).

Undergraduate Thesis, Dept. of CE, BUET 2023 – 2024

Supervisor: Prof. Mohammed Kabirul Islam

Experimental and Numerical Investigation of Bearing Capacity of Circular Footing on Clay Soil

Investigated the bearing behavior of circular footings on clay through laboratory experiments and validated the results with finite element analysis in PLAXIS 3D.

- Designed the experimental setup accounting for boundary effects.
- Analyzed load–settlement response and the relationship between settlement and footing diameter, comparing experimental and numerical results.

PROFESSIONAL EXPERIENCE

Graduate Engineer Dec 2024 – May 2025

GeoStruct Consultants Ltd. Dhaka, Bangladesh

- **Remedial Measures to Strengthen Salehpur Bridge, Amin Bazar, Dhaka**: Modeled and analyzed a 51 m steel composite bridge in Midas Civil, applying AASHTO, BNBC, AISC, and ACI for design verification; coordinated with multidisciplinary teams on design outputs and construction feasibility.
- Prepared Bills of Quantities (BOQ), detailed engineering drawings, technical reports, and presentations for several other projects; conducted research to support design decisions.

CONFERENCES AND SITE VISITS

- **Presenter**, 7th International Conference on Advances in Civil Engineering (ICACE), CUET Dec 2024
- **Site Visit & Seminar**, Hazrat Shahjalal International Airport Terminal 3 Project, Dhaka
- **Factory Visit**, Hatim Concrete Industries Ltd., Rupganj, Narayanganj

HONORS AND AWARDS

- **University Technical Scholarship (BUET)** 2019 – 2023
- **2nd Runner-Up**, ACI BUET Student Chapter Poster Competition (Recycled Concrete using Nano Silica) 2022
- **Participation Award**, CEismic Poster Competition, BUET (Response of Underground Tunnel to Combined Dynamic Loading of Earthquake and Running Train) 2023
- **Achievement Award (7th)**, BAS Divisional Science Olympiad, Bangladesh Academy of Sciences 2015

SELECTED PROJECTS

Capstone: Development of a Multi-Storied Female Dormitory at Aftabnagar June 2023 – Feb 2024

A comprehensive feasibility study assessing the technical, financial, and economic viability of a multi-storied dormitory in Aftabnagar, Bangladesh, addressing complex engineering problems including foundation system design and structural system functionality.

SKILLS

Software	PLAXIS 2D/3D, Midas Civil, ETABS, SAP2000, AutoCAD, HighScore Plus
Laboratory Testing	Grain size analysis (sieve & hydrometer), Atterberg limits, Specific gravity, Compaction, Unconfined compression, Direct shear, 1D consolidation, Free swell, Swell pressure, Dispersivity (Pinhole, Crumb, Double hydrometer), ESP, SEM, XRD
Programming	C++
Tools	Microsoft Office, Affinity Designer, Adobe Creative Suite, Primavera
Soft Skills	Project Management, Communication, Team Building, Technical Writing

AFFILIATIONS AND EXTRA-CURRICULAR

- **General Secretary & Founding Member**, ASCE Student Chapter, BUET Aug 2022 – June 2024
- **Director of Publications**, BUET Photographic Society June 2022 – Sept 2024
- **Assistant Secretary (Research & Competition)**, BUET Journalism Society Apr 2021 – 2022

REFERENCES

Dr. Mohammad Shariful Islam
Professor, Dept. of CE, BUET
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Dr. Mohammed Kabirul Islam
Professor, Dept. of CE, BUET
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Dr. Mahbuba Begum
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